Alpha Diversity

Muluh

Community diversity

Group-wise comparisons

- **Index:** observed
- **Group variable:** diet
- **Multiple testing correction:** fdr
- **Statistical test** Wilcoxon

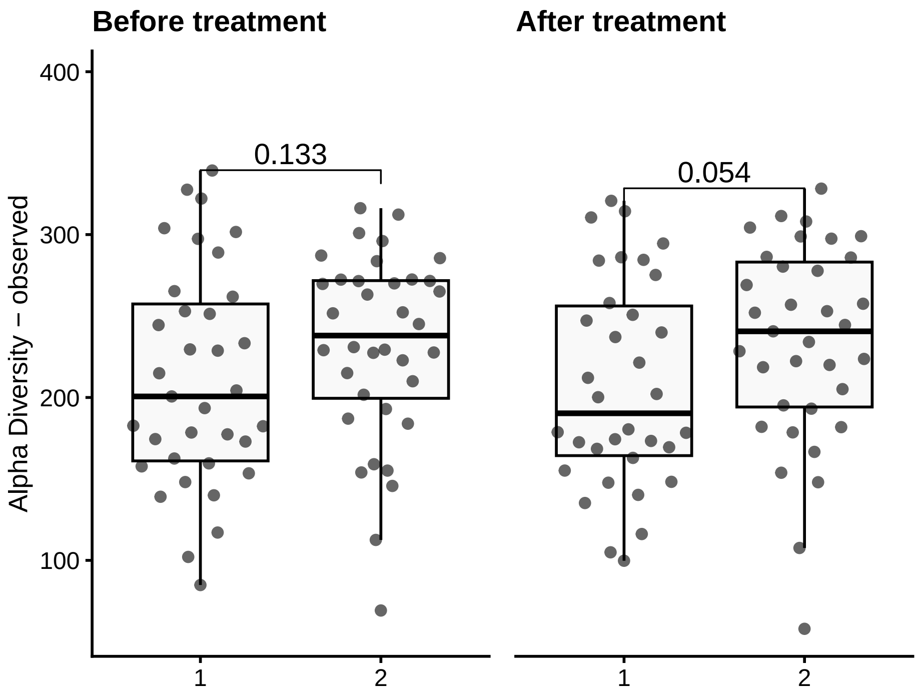


Table of the top findings

Across-group

Before treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

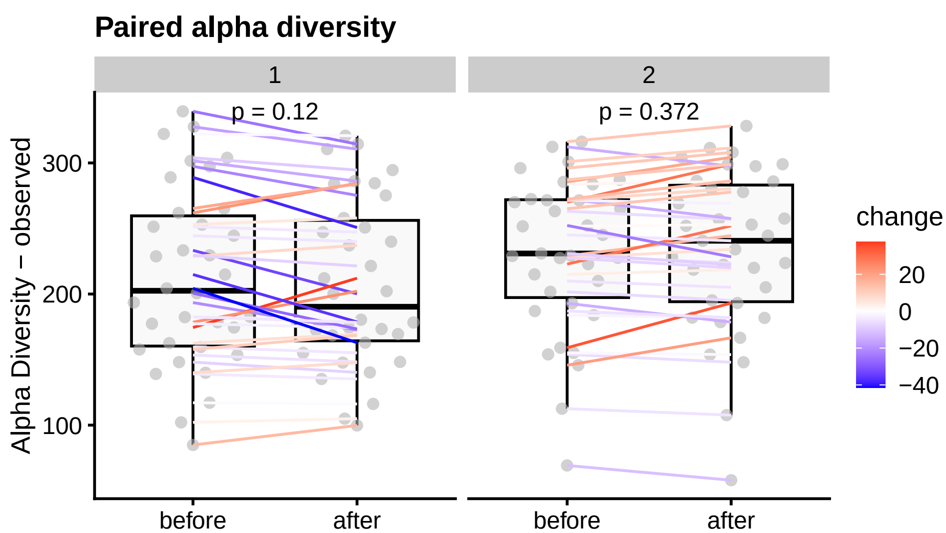
group1	group2	mean1	mean2	log2FC	p.adj
1	2	211.3	231.7	0.13	0.133

After treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	207.2	233.4	0.17	0.054



Within-group

Characteristic	before			after		
	Beta	95% CI[1]	p-value	Beta	95% CI[1]	p-value
(Intercept)	151	67, 236	<0.001	144	59, 229	0.001
meal_group			0.4			0.3
1	—	—		—	—	
2	32	-11, 75	0.14	28	-16, 72	0.2
3	28	-15, 71	0.2	30	-14, 73	0.2
4	10	-32, 53	0.6	-3.9	-47, 39	0.9
age	1.0	-0.44, 2.5	0.2	1.2	-0.25, 2.7	0.10

[1]CI = Confidence Interval

Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

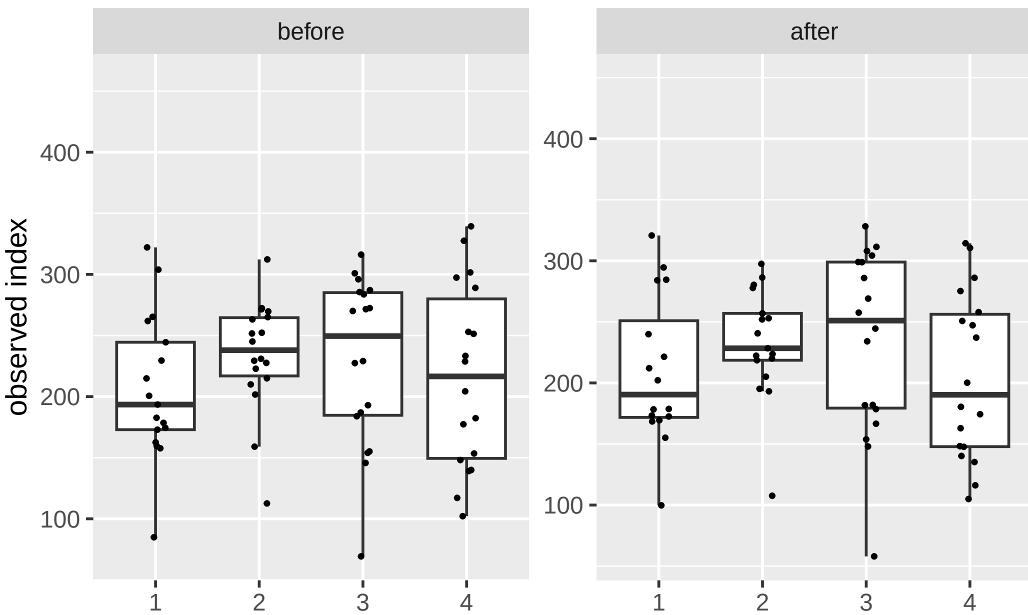
diet	mean1	mean2	log2FC	p.value
1	211.3	207.2	0.04	0.120
2	231.7	233.4	0.13	0.372

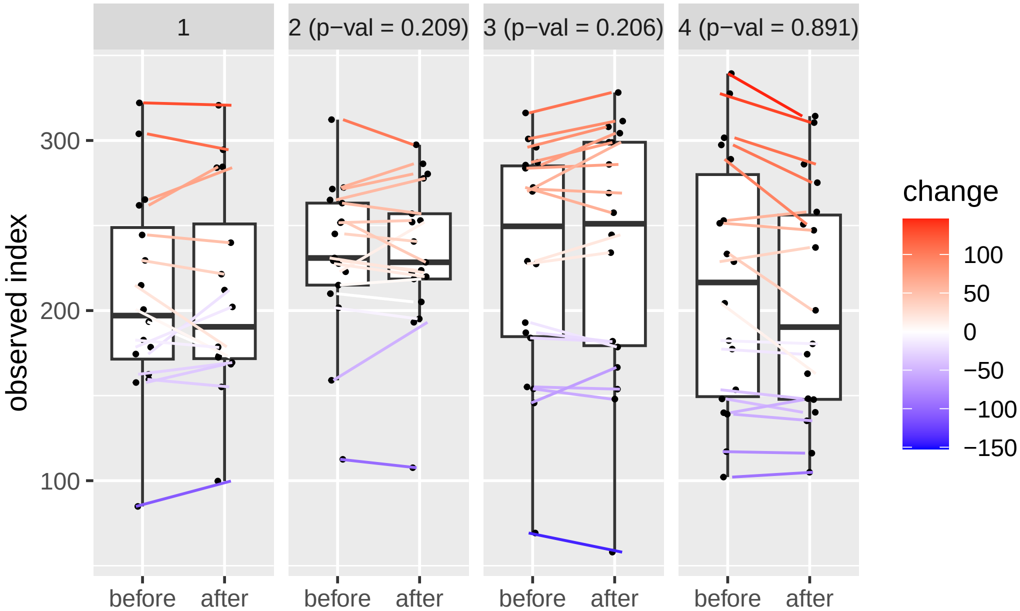
Alternate analyses

- **Beta (Estimate)**: Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI**: Range where the true beta is likely to fall, reflecting precision.
- **p-Value**: Tests if beta differs from zero; p<0.05p < 0.05p<0.05 suggests significance.

Paired test between timepoints (1 and 2)	Beta	95% CI[1]	p-value
(Intercept)	147	84, 210	<0.001
meal_group			0.6
1	—	—	
2	28	-16, 71	0.2
3	28	-15, 71	0.2
4	2.1	-28, 32	0.9
age	1.2	0.14, 2.2	0.025

[1]CI = Confidence Interval





Data Summary

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class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(14): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(13): PrevalentGenus PrevalentSpecies ... genus_prevalent
  pathabundance
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```